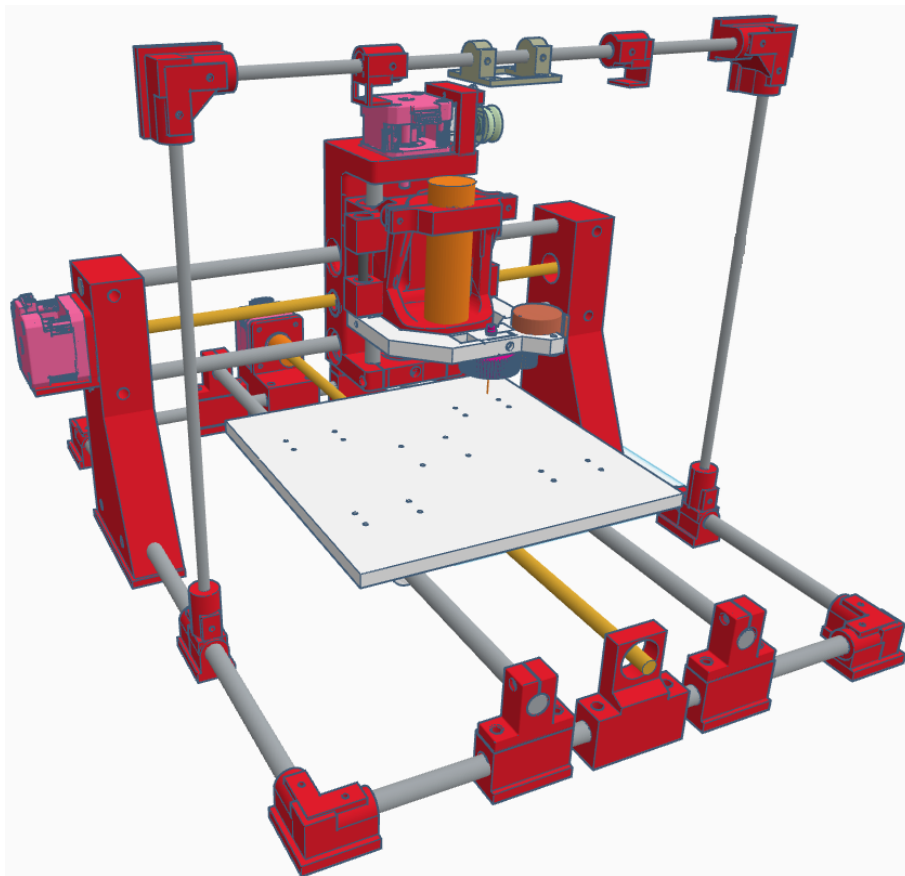


About This Manual

In this manual I'd like to create a step-by-step tutorial that will guide you through building your own LPODRO machine.

This is the first version of this manual that is still a WIP. In the future I'd like it to be in a easy to read form, with a lot of pictures that will show what exactly you need to do. I also plan to record a video tutorial to further simplify the building process.

As of right now the current state of this manual is more like an info dump by me – sharing everything I learned over the past 3 years while working on this project. Most likely I will skip or forget something here, that's why in the readme file it says that you need to have at least some knowledge of how CNC machines are built. At this point it is more than likely that you will need to figure out some stuff out on your own.



This is the visualization picture. While following this manual keep in mind that this ^ is what you are building. Each step will get you closer to having this machine. After each step check if whatever you did looks like what is on this picture. If not you most likely did something wrong.

Frame Assembly

Step 1: Cut the rods to length

To start out the building process you need to cut the 10mm rod into the proper lengths: 2x 41cm and 2x 31.5cm for the main frame, 2x 44cm for the y axis, 2x 36.5cm for the x axis and 2x 14cm for the z axis. For this I used an angle grinder. I also used it to round out the rough edges. Be aware that the rods will get very hot – don't touch them and don't put them into the 3d printed parts right after cutting them.

Step 2: Check the 3D printed parts fit the rods

After cutting the rod, take the ones that will be used for building the main frame alongside 4 angle connectors, both X axis pillars, 4 Y axis rod holders, Y axis bearing holder, Y axis stepper holder and 2 camera frame holders. Take the 3d printed parts and check if the hole that the rods will go into have the proper diameter. To do so take a rod and try inserting it into the hole. There shouldn't be too much resistance (if you put too much pressure the part will just split open on the layer lines) but also the rod cannot be extremely loose. It should go in with just a bit of pressure and have a bit of give in each direction. If it doesn't either drill the hole out to a proper size or file it down. Now you need to make a choice because you can either put threaded inserts into the 3d printed parts or screw a screw right into it. Personally I did the latter because it's much simpler and it's strong enough for what you're doing here. But if you insist on using threaded inserts the tip I can give you is after melting it in wait a bit to let the molten plastic harden. When it does take something thin, that can be inserted into the threaded insert without touching the threads, put it as far as you can and hit it pretty hard with your hand or a hammer. This will break out the plastic that has molten into the part that the rod will go into. If you don't do this you won't be able to screw the screw or put the rod in. If you choose to screw into the parts you need to do this at least once before putting the rod in.

Step 3: Insert the rods into the connectors

Next, take the two shorter rods, insert the Y axis rod holder, Y axis bearing holder and again a Y axis rod holder onto the first and and Y axis rod holder, Y axis stepper holder and again a Y axis rod holder on the second one. Insert each of the X axis pillars and camera frame connectors onto the longer rods.

Step 4: Form and square the main frame

After that insert the prepared rods into the angle connectors to form the main frame with inserted components in such a way that each part is orientated as shown on the visualization picture. If what you have looks similarly to the visualization picture take 8 screws and put them into the angle connectors in such a way that they won't fall out but won't stop the rod from moving freely in the connectors. Now you need to tighten the screws but also at the same time make sure that the rods form a right angle with each other to get a square frame. To do so you need to grab a machinist's square, put it in such a way that one of its arms goes parallel to one rod and the second one with a second and when the rods form a right angle tighten the screw. Repeat this process for all corners.

Bed Assembly

Step 5: Prepare the bed plate and nut holders

With that done the next step is preparing the bed. Personally I ordered it to be cut out from PMMA but if not for the fact that I already had a PMMA sheet lying around that would have been very expensive. What I recommend doing is to buy a 25x25cm, 4mm thick steel sheet, drill the holes and make the camera cutout according to the bed.svg drawing and tap the holes with an m3 thread. Next take 4 bed bearing holders and both nut holders. Take the first T8 nut and screw it completely into the holder with holes. The second T8 nut goes into the holder with slots, put the 4 screws in, screw the m3 nuts onto them but don't tighten them – the nut needs to turn freely for now. Next take the bearings and push them into the holders. It should require a lot of force so they don't fall out during normal operation but if you see any cracks forming on the holders during this step, replace the cracked part with a new one and file everything down a bit. After inserting the bearings make sure that one side is absolutely flush with its holder. It cannot be crooked or anything like that.

Important: Pay EXTRA attention to the bearings being flush with their holders – I wasted literal days trying to fix my machine only to find out I installed one of the bearings at an angle.

With that finished, take the bed and the prepared holders and screw them together. Make sure that the screws don't poke out on the other side of the bed. If they do, add a nut or a washer to the screws.

Step 6: Mount the bed camera

As the last step take the bed camera (the ov9726 one with the lens from the second one), place it under the bed so that the lens points up and screw it in from the top.

Checkpoint: *You should have a fully assembled bed with one loose T8 nut.*

Y Axis Assembly

Step 7: Prepare the Y axis stepper motor and bearing

Now you can take the Y axis stepper motor, a coupler and a T8 screw. Put the coupler onto the motor shaft and screw it in in such a way that one of the coupler screws pushes onto the flat part of the shaft. Next take the T8 screw, put it in the coupler and slowly tighten one of the coupler screws while rotating the T8 screw – the goal of this is to screw the coupler screw into the thread of the T8 screw. You will feel it when you hit the right spot. When you do it, tighten that coupler screw fully and then the second one and put this contraption aside. Now take the 608zz bearing and install it into the Y axis bearing holder.

Step 8: Slide the bed onto the Y axis rods

After that take the Y axis rods and put them through the first holder but not the second – leave the ends of the rods floating approximately in the middle, then with one hand hold the ends of the rods that are outside the main frame and with the second one pick up the bed and slide it onto the rods so that it will be inside the main frame. After that slide the rods fully in so that they go into the second rod holders.

Checkpoint: *You should have the main frame with a bed that can freely slide back and forth and be moved on the frame.*

Step 9: Install the Y axis motor and center the bed

Now take the stepper motor with the T8 screw, put the screw through the Y axis motor mount and push it until the screws tip touches the first nut. When it does start turning the screw so that it screws into the first nut. Do it until the tip of the T8 screw reaches the second, loose nut. When it does you'll most likely need to turn the nut a bit to let the screw screw into it. When it does turn the screw until the stepper motor touches the Y axis motor mount and the tip of the T8 screw goes through the Y axis bearing. With that done you can screw the motor to its mount and tighten the rod mounts. After that we need to make sure that the bed is centered before screwing all of its mounts to the frame. To do so turn the T8 screw manually until the bed reaches one of the ends of the frame, then take some sort of measuring device and measure the distance between the rod holders and the frame rods. If the distances are not equal, move the holders a bit and repeat the process until they are. When they are equal, tighten the screw on the two holders on the side on which the bed is, then move the middle component (either the bearing or the stepper mount) and see if it springs back to a certain place (should be the middle between both rod holders), if it does, screw it in, if it doesn't manually place it in the middle and then screw it in. After that manually move the bed to the other side and repeat this process. After that's done make sure that when rotating the T8 screw manually the resistance you feel is equal along the whole length. If it doesn't it means that something isn't parallel to each other. Check the rod holder, stepper and bearing mounts distance to each other and the frame and also if the rods and the T8 screw you installed isn't bend. Now you need to properly tighten the loose T8 nut. What you are trying to achieve is to eliminate any play in the bed movement without adding any additional resistance for the stepper motor. To do so, grab both sides of the bed with your hands and try moving it left and right (in the direction of the rods and T8 screw). The nut should be held tight in its holder that it shouldn't move on its own. If you feel a bit of give caused by improper thread spacing (what we are trying to eliminate right now), and not because something else is moving (the frame, stepper mount, the loose nut in its holder etc), turn the nut a bit and try again. When the give stops try manually turning the T8 screw. If you feel a lot of additional resistance it means that you turned the nut too much. Now you need to find an angle where the bed doesn't have any give but also when turning the T8 screw you don't feel any additional resistance.

When you find this point screw the 4 screw on the nut all the way in.

Checkpoint: *You should have the main frame with a bed that should smoothly move back and forth when rotating the T8 screw but should not move at all in any direction when pushing on the bed itself.*

PnP Head Assembly

Step 10: Assemble the PnP head

Now you are going to assemble the PnP head as it will make the next step easier. To do so take the PnP head frame, gear 1 and gear 2. Then insert 3 m3 nuts into the PnP head frame - two in the designated cuts next to the rod holes and one in the cut next to the gear 1 hole. Next take the 28byj-48 stepper motor and push the gear 2 onto its shaft. After that put it in the PnP head frame and screw it in. Next take the bearing and push it onto its place. With that done push the clear tube through the top of the PnP head frame. From the bottom push the gear 1 into the tube and then pull the tube back so that the part of the gear 1 that is in it goes through the hole of the PnP head frame. With that placed correctly take an m3 screw and screw it in the hole next to the gear 1. Tighten it until the tube and gear 1 connection is airtight but don't over do it so that the stepper motor can still rotate the gear 1. Next take the properly cut rods, insert them into their holes, make sure they form right angles with the PnP head frame and tighten the screws.

Spindle Carriage Assembly

Step 11: Glue the Z axis bearing holders (if using the standalone carriage)

Now you're going to assemble the spindle carriage. To do so you can either use the stand alone spindle carriage and glue the Z axis bearing holders onto it or the complete spindle carriage. Personally I chose the first option because it was simpler to print and more forgiving on my not so good 3d printer tolerances. But what you choose is completely up to you - the complete carriage has the advantage of not needing any gluing. If you choose to glue the bearing holders onto the carriage the PnP head that you just assembled is going to help you. To do it push the bearings into the holders (remember, they need to be PERFECTLY flush!) then slide them onto the rods of the PnP head, then apply a bit of super glue onto the contact area of the bearing holders and slide them into the proper height. You need to do bearings from both sides at the same time. While doing so make sure that the rod goes parallel with the spindle carriage and that it's not crooked. When the glue dries a bit you can glue the remaining two bearing holders.

Step 12: Install bearings, T8 nut, and fit the spindle

Next take the spindle carriage (either the complete one or the glued one) and push 2 or 4 bearings into the bearing holes (once again its VERY IMPORTANT that they are COMPLETELY flat). I recommend using 4 for better stability and longevity. Next push one T8 nut and screw it in completely on one side and on the other side do the same eliminating the play trick you used on the bed. The last step with the spindle carriage is to sand down the part that actually touches the spindle. To do so take the spindle clamp and the spindle itself. Unscrew the top cap of the spindle and push the spindle clamp onto it then screw the cap back on. Then put the spindle with the clamp into the carriage and screw the two screws completely in. From the front the spindle should go completely down, if its at an angle something's really wrong - most likely the 3d print quality is bad (take note of that as this can cause A LOT of different problems). From the side tho it will most likely be at an angle - that is to be expected. If it is look from the top, maybe shine a light from the bottom. You're trying to see what parts actually touch the spindle. When you do find them take the spindle out, sand those parts a bit, put the spindle back in and check from the side if it points straight down. If it doesn't repeat this process until it does.

Checkpoint: *When it finally does, congrats – you have a finished spindle carriage.*

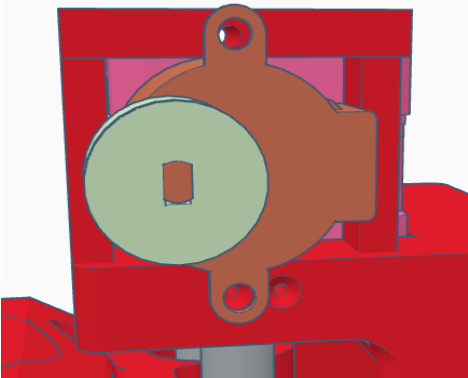
Z Axis and X Axis Assembly

Step 13: Build the Z axis carriage

With the finished spindle carriage we can assemble the X axis. To do so take the Z axis carriage and the second Z axis stepper holder and spool. Insert either 2 or 4 bearings into the Z axis carriage (once again they need to be PERFECTLY flush and straight) and two T8 nuts and do the eliminating the play thing - just like with the spindle carriage. Next push a bearing into the bottom part of the Z axis carriage. Next take a Nema17 stepper motor and install a coupler and a T8 screw just like you did with the Y axis. Then insert it through the top of the z axis carriage until the tip of the T8 screw goes into the bearing. The face of the nema17 has to be flush with its designated place on the z axis carriage. If it isn't because the T8 screw is slightly longer you can drill a hole under the bearing but be aware that the T8 screw cant poke under the Z axis carriage. If the motor sits flush with the z axis carriage and the T8 screw doesn't stick out of the bottom you can slightly rise the nema17 until you can put the spindle carriage under the tip of the T8 screw. Next screw the screw into the spindle carriage. With that done screw the nema 17 into the Z axis carriage. Next push two rods in and make sure they go into the holes on the top of the Z axis carriage. After that make sure that they don't poke out from the bottom. If they don't you can screw an m3 screw into their designated spots (no nut here, it screws into the 3d print itself). That should lock the rods in place.

Step 14: Attach the second Z axis stepper, pump, and string spool

The next step would be to glue the second Z axis stepper holder and the onto the top the Z axis carriage. To do so take a 28byj-48 stepper and push the second Z axis spool onto its shaft. Then insert the motor into the holder as shown on the picture, screw it together, place it on top of the z axis carriage and screw the second screw to it. With that done put some super glue along side the edge of the connection between the Z axis carriage and the second Z axis stepper holder and let it dry. To glue to pump in place just place it where it should be and put a lot of hotglue there. When it dries off connect the tube from the PnP head to the pump. To finish everything off take a piece of string, put a small dab of super glue in the middle of the spool, put the end of the string into that dab, wait for it to dry, when it does spool some string onto it, then take the string down and cut it so that it can reach to the bottom



position of the pnp head (if the pnp head isn't inserted into the spindle holder that's a good time to do it). Then tie that end of the string onto the screw that is tightening the rod that's closer to the spooled stepper to the pnp head frame. With that done turn the spooled stepper so that the pnp head goes as far as it can, take a limit switch, push it into the pnp head until it clicks, then push it a bit more and glue it to one of the second Z axis bottom bearing holders.

Checkpoint: You should have a complete carriage that holds the spindle, PnP head and the pressure pump that will now go onto the X axis.

Step 15: Mount the assembly onto the X axis

To do so take a properly cut rod, insert it partially into the top hole of one of the X axis pillars and push it in until the tip reaches about half way. Then push the X axis assembly onto it and push the rod in until it goes into the hole of the second X axis pillar. Then push the second rod all the way through, assemble another nema17, coupler, T8 screw combo and push the T8 screw through the nema17 hole until its tip reaches the first nut, then screw it in completely and when the nema17 sits flush with the pillar screw it in. Lastly insert the 608zz bearing into the second x axis pillar, the one that houses the T8 screw tip

Step 16: Verify the mechanical assembly

With that done you should have an almost complete LPODRO machine. The only things that are missing are the camera assembly, the camera itself and the limit switches. Before we can go further you need to check if everything that you have build so far is correct. Check if it looks like in the visualization picture, check every angle you can imagine (if all of the frame corners are right angles, does the X pillars form right angles etc.), check if while moving the X, Y or Z axis manually you don't encounter extra resistance somewhere along its movement path, check if during the assembly any screw didn't come loose.

Camera Frame Assembly

Step 17: Build and align the camera frame

If everything checks out you can assemble the camera frame. To do so take two angle connector and a proper camera holder - either camera holder or phone holder 1 and 2. If you use the normal camera holder this is a good time to screw the camera into it. Next take the rod, slide the camera connectors onto it and at each end of that rod place the angle connectors. Next place the properly cut rods into the camera frame connector and insert into those rods that angle connectors with the rod and camera holder. Now you should have a camera frame but it still needs to be properly placed, to do so move the bed into the proper position (the best way to do this is through the calibration menu from the python program but to use that you need to first complete this guide and then come back to this step), and move the whole camera assembly until the you see the whole bed on the camera feed. With that done you can screw/tighten all the screw while checking for right angles in the whole frame.

Limit Switches

Step 18: Install the limit switches

Now the last remaining step is to install the limit switches. Right now I don't have a way to screw them in or something - I just glued them. To do that correctly move for the axis (so the bed, z axis carriage or the spindle carriage) the the minimal position it can go to (looking from the front the bed has to go to the back, the z axis carriage to the left and spindle carriage to the top), then push a switch up against it until it clicks, then push it a bit more and then glue it in place - the exact same process as with the pnp head.

***Checkpoint:** This was the last step of the mechanical assembly of LPODRO. At this point you should have something that looks exactly like on the visualization photo.*

Electrical Assembly

What you need to do now is the electrical assembly (it mainly consists of just connecting cables to different places). What I can only say here is that everything that's needed is on the schematic, the example values can be found in the BOM, make sure that your cnc shield is not a cheap clone because if it is there is most likely a mistake in it - google cnc shield v4 fix.

Important – remove the diode from the CNC shield: If you won't you won't be able to power the stepper motors with 24v and you will have to use 12v instead. Also it improves safety. If you're not interested as to why, just skip to the next section. If you are, here you go: that diode connects the DC jack on the CNC shield to the VIN pin on the Arduino Nano. This means that the Arduino is powered more from the 12v than from the USB cable. This is a problem because if you short out the Arduino Nano (which is surprisingly easy) you will, best case scenario, blow out the 5v regulator on the Arduino Nano – but in my experience that happened only once out of 6 times. The first 2 shorts caused the atmega chip to fry itself, the third time damaged the 5v regulator but in a really weird way, because if it would have been powered with up to 10v everything would work perfectly fine, but if that regulator would get anything higher than 10v (which is what it would get if that diode had not been removed) the Arduino would go absolutely bananas. The next 2

shorts caused partial damage to the atmega chip and the last time fried the diode that's on the Arduino Nano. If you remove the diode from the CNC shield, the Arduino Nano will need to be powered from USB. And that is good, because the USB port on laptops but also on the Raspberry Pi Zero W has a 500mA resettable fuse, so if you short something out now the Arduino will lose power long before any permanent damage can occur. Also by removing this diode you remove the only 12v part from this line and this allows for the use of 24v which allows to run the steppers at a higher torque.

After the electrical assembly the last step of this manual is the first configuration - stuff you need to do only once.

GRBL Configuration

Step 19: Flash the Arduino Nano with GRBL

First order of business would be to flash the Arduino Nano with GRBL. This is a good tutorial: <https://www.instructables.com/How-to-Install-use-GRBL-With-Your-Cnc-Machine/> – just don't use the Prusa steps calculator. Instead, for the whole config, use these values:

\$0 = 10 (Step pulse time, microseconds)
\$1 = 255 (Step idle delay, milliseconds)
\$2 = 0 (Step pulse invert, mask)
\$3 = 4 (Step direction invert, mask)
\$4 = 0 (Invert step enable pin, boolean)
\$5 = 1 (Invert limit pins, boolean)
\$6 = 0 (Invert probe pin, boolean)
\$10 = 1 (Status report options, mask)
\$11 = 0.010 (Junction deviation, millimeters)
\$12 = 0.002 (Arc tolerance, millimeters)
\$13 = 0 (Report in inches, boolean)
\$20 = 0 (Soft limits enable, boolean)
\$21 = 1 (Hard limits enable, boolean)
\$22 = 1 (Homing cycle enable, boolean)
\$23 = 3 (Homing direction invert, mask)
\$24 = 50.000 (Homing locate feed rate, mm/min)
\$25 = 500.000 (Homing search seek rate, mm/min)
\$26 = 250 (Homing switch debounce delay, milliseconds)
\$27 = 3.000 (Homing switch pull-off distance, millimeters)
\$30 = 1000 (Maximum spindle speed, RPM)
\$31 = 0 (Minimum spindle speed, RPM)
\$32 = 0 (Laser-mode enable, boolean)
\$100 = 400.000 (X-axis travel resolution, step/mm)
\$101 = 400.000 (Y-axis travel resolution, step/mm)
\$102 = 400.000 (Z-axis travel resolution, step/mm)
\$110 = 500.000 (X-axis maximum rate, mm/min)
\$111 = 500.000 (Y-axis maximum rate, mm/min)
\$112 = 50.000 (Z-axis maximum rate, mm/min)
\$120 = 800.000 (X-axis acceleration, mm/sec²)
\$121 = 800.000 (Y-axis acceleration, mm/sec²)
\$122 = 100.000 (Z-axis acceleration, mm/sec²)
\$130 = 300.000 (X-axis maximum travel, millimeters)
\$131 = 300.000 (Y-axis maximum travel, millimeters)
\$132 = 50.000 (Z-axis maximum travel, millimeters)

Step 20: Calculate your steps/mm

For the steps per mm you need to calculate your own values. The formula is as follows: $360 : \text{step angle} : T8 \text{ thread pitch} * \text{your microsteps setting}$. For example, my stepper motor has a step angle of

1.8 degrees, my T8 screw has an 8mm thread pitch and I set the shield to 1/16 of a step, so putting that together it's $360:1.8:8*16$ which comes out to 400. After that, put the Arduino into the CNC shield.

Raspberry Pi and Software Setup

Step 21: Flash and configure the Raspberry Pi

Next you'll take care of the Raspberry Pi Zero W. Flash the SD card with the newest Raspberry Pi OS Lite image (make sure to set up automatic WiFi connection), boot it up, and SSH into it, then update the system, install Python 3.14 and pip. Then over FTP or SCP transfer the requirement.txt and rpi.py file onto the Pi, install the needed libraries using pip, and set the OS so that rpi.py launches on startup.

Step 22: Set up the control software on your PC

Next, on the support PC (the one that will be connected to LPODRO), also install libraries using pip, install pcb2gcode and launch the main.py file. During the first boot you'll get a warning about a missing config. Press enter, a new config file has been created automatically and has everything you'll need with default values that are a good starting point. Next you need to configure the connection with the Arduino Nano and the cameras, just press 4 in the main menu and connect to everything. When you're finished, quit the app and relaunch it, check if the connection settings saved and when you press yes to reconnect check if the script connected with the Raspberry Pi (the PC needs to be on the same network as the Rpi!).

Step 23: Mark the bed

With the control software set up you can now precisely move the machine around using UGS. With that you can precisely move the camera (you can go back to that step right now) and more importantly mark the bed. This is later used by the python program to detect the exact location of stuff placed on the bed. To mark the bed, home the machine and manually move it in such a way that the tip of the mill will be at the bottom left corner of the bed. This is your 0,0 point. Now move to 0,0 0,194 194,194 and 194,0 and at each of those positions mill a small hole in the bed. With that done clean them of any debris and put a small amount of paint in there. I used some nail paint.

You're Done

If it does, congrats! You have a fully assembled and configured LPODRO machine that can be used to make PCBs! Now it is probably best to head to the user manual and try making your first PCB!